

KATHMANDU UNIVERSITY
End Semester Examination
March/April, 2025

Marks Scored:

Level : B.E./BIT
Year : I

Course : MATH 101
Semester : I

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No.:

Date : 02 APR 2025

SECTION "A"
[10 Q. $\times$ 1 = 10 marks]

Fill in the blank space(s) by writing the most appropriate word(s) or symbol(s).

1. The domain of f is A and domain of g is B then the domain of fg is _____.
2. $\lim_{x \rightarrow \frac{5}{2}} \left[\frac{x}{2} \right] = \underline{\hspace{2cm}}$ where $[]$ denote it's usual meaning
3. The function $y = |x|$ is differentiable at every point of x except _____.
4. The column of a matrix A are linearly independent if and only if the echelon form of A has _____.
5. The vertical asymptotes of the function $y = \frac{2x}{x-3}$ is _____.
6. The critical point of $y = x^2 - 2x$ is _____.
7. If the series is convergent and whose sum of the n -th term is $y = \frac{2n}{3n+5}$, then the limit is _____.
8. If a matrix A has n columns, then $\text{rank } A + \dim(\text{Null } A) = \underline{\hspace{2cm}}$.
9. The scalar 7 is an eigenvalue of A , if and only if the equation $Ax = \underline{\hspace{2cm}}$.
10. If a set contains more vectors than there are entries in each vector, then the set is _____.

SECTION "B"

[10 Q.× 1 = 10 marks]

Fill in the blank space(s), **DO NOT TICK**, by selecting the most appropriate answers from among the given ones.

11. If $f(x) = -\sqrt{x}$ shifted right by 3 units, then the new function is given by _____.
 $[-\sqrt{x-3}, \quad \sqrt{x-3}, \quad \sqrt{x+3}, \quad -\sqrt{x+3}]$
12. If f' changes from positive to negative at c , then f has: _____ at c
 $[\text{Local maxima}, \quad \text{Local minima}, \quad \text{Absolute minima}, \quad \text{No extreme values}]$
13. The interval in which function $f(x) = x^2 - 4x + 5$ is increasing is _____.
 $[(0, \infty), \quad (2, \infty), \quad (-\infty, 2), \quad (4, \infty)]$
14. At $x=1$, the graph of the function $y = 4x^2 - 2x$ is _____.
 $[\text{Concave up}, \quad \text{Concave down}, \quad \text{Stationary}, \quad \text{Decreasing}]$
15. Which of the following is the same as the definite integral $\int_a^b f(x) dx$: _____.
 $[\int_a^b f(x) da, \quad \int_a^b f(x) db, \quad -\int_b^a f(x) dx, \quad -\int_a^b f(x) dx]$
16. The improper integral $\int_0^\infty \frac{dx}{x^2+4}$: _____.
 $[\text{Diverges to infinity}, \quad \text{Converges to zero}, \quad \text{Converges to } \frac{\pi}{4}, \quad \text{Converges to } \frac{\pi}{2}]$
17. The series $\sum n^{-p}$ is convergent iff _____.
 $[p < 1, \quad p > 1, \quad p = 1, \quad p \leq 1]$
18. If one row in an echelon form of an augmented matrix is $[0 \ 0 \ 0 \ 5]$ then the associated linear system has _____.
 $[\text{exactly one solution}, \quad \text{two solution}, \quad \text{infinitely many solution}, \quad \text{no solution}]$
19. The maximum rank of a 3×3 matrix is: _____.
 $[0, \quad 1, \quad 2, \quad 3]$
20. The scalar λ is a characteristic root of the matrix A if _____ is singular
 $[A, \quad \lambda I, \quad \lambda - AI, \quad A - \lambda I]$